

UNIT I – Lecture Module 2

CHARACTERISTICS OF CLOUD COMPUTING**1. INTRODUCTION:**

Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction.

The National Institute of Standards and Technology (NIST) — in Special Publication 800-145 — defines cloud computing through five essential characteristics, three service models, and four deployment models. Understanding the characteristics is fundamental to appreciating why cloud computing represents a paradigm shift over traditional IT infrastructure.

 NIST DEFINITION FRAMEWORK (SP 800-145)

Component	Elements
Essential Characteristics	On-Demand Self-Service • Broad Network Access • Resource Pooling • Rapid Elasticity • Measured Service
Service Models	Infrastructure as a Service (IaaS) • Platform as a Service (PaaS) • Software as a Service (SaaS)
Deployment Models	Public Cloud • Private Cloud • Community Cloud • Hybrid Cloud

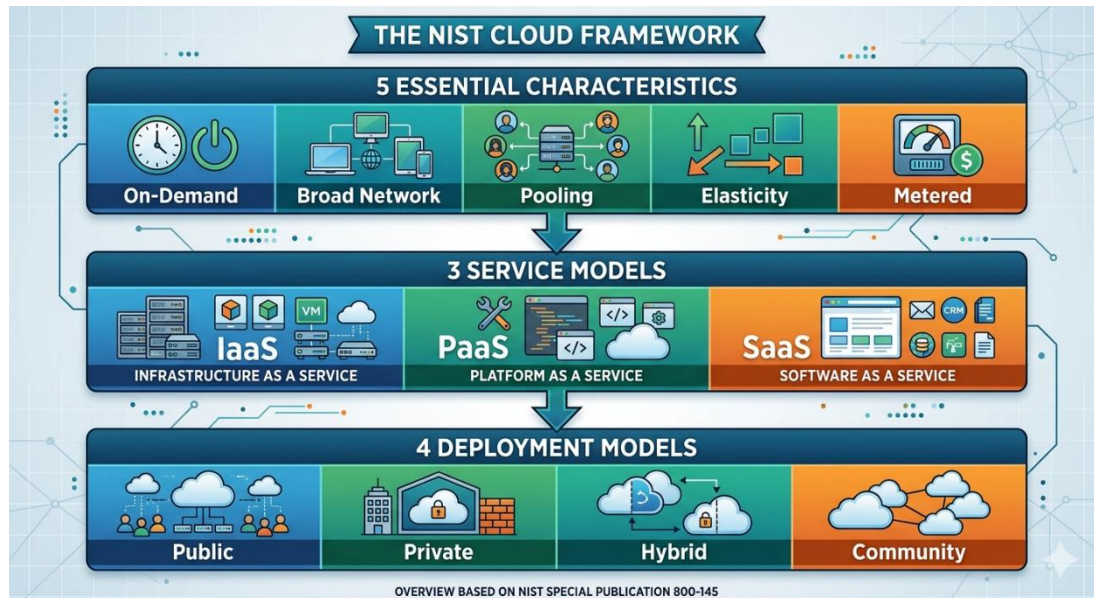
THE NIST CONCEPTUAL FRAMEWORK:

To establish a standard industry terminology, the **National Institute of Standards and Technology (NIST)** developed a formal, universally accepted definition of cloud computing (*NIST Special Publication 800-145*).

NIST:

"Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction."

The NIST framework categorizes cloud computing into three distinct architectural layers: **5 Essential Characteristics**, **3 Service Models**, and **4 Deployment Models**.



THE FIVE ESSENTIAL CHARACTERISTICS:

For a service to be considered cloud computing, it must exhibit all five essential characteristics defined by NIST. These characteristics collectively distinguish cloud computing from conventional hosting, virtualization, or outsourcing.

1. ON-DEMAND SELF-SERVICE:

A consumer can unilaterally provision computing capabilities, such as server time and network storage, as needed automatically without requiring human interaction with each service provider.

1. On-Demand Self-Service

Traditional IT provisioning requires the user to submit a formal request, which is then reviewed, approved, and manually fulfilled by the IT department — a process that can take days or weeks. Cloud computing eliminates this bottleneck entirely.

Key aspects of on-demand self-service:

- **Self-provisioning:** Provisioning is automated through web-based self-service portals or APIs.
- **Always-available:** Resources are available 24×7 without scheduled maintenance windows.
- **No intermediary:** No human approval workflow is needed for standard resource requests.
- **Examples:** AWS Management Console, Azure Portal, and Google Cloud Console are examples of self-service interfaces.

Real-World Example: A developer can launch a new virtual machine on AWS EC2 within 60 seconds using the web console, without ever contacting a system administrator.

2.BROAD NETWORK ACCESS:

Capabilities are available over the network and accessed through standard mechanisms that promote use by heterogeneous thin or thick client platforms (e.g., mobile phones, tablets, laptops, and workstations).

Broad Network Access – Explained

Cloud resources are delivered over standard Internet protocols (HTTP/HTTPS, REST APIs, etc.) and are accessible from any device with a network connection. This is in contrast to traditional applications that were often tied to specific hardware or local area networks.

- **Network delivery:** Services are reachable via public Internet or dedicated private connections (e.g., AWS Direct Connect, Azure ExpressRoute).
- **Platform-agnostic:** Access from desktops, laptops, tablets, smartphones, and IoT devices using standard browsers or client apps.
- **Thin & Thick clients:** Both thin clients (web browsers) and thick clients (native apps) are supported.
- **Interoperability:** Standardized protocols ensure interoperability across vendors and platforms.

Access Type	Description	Example
Web Browser (Thin)	No local installation required	Gmail, Google Docs
Native App (Thick)	Installed client with richer features	Microsoft 365 Desktop App
API/CLI Access	Programmatic access for automation	AWS CLI, Azure SDK
Mobile App	Access from smartphones/tablets	Dropbox Mobile App

3.RESOURCE POOLING:

The provider's computing resources are pooled to serve multiple consumers using a multi-tenant model, with different physical and virtual resources dynamically assigned and reassigned according to consumer demand.

Resource Pooling – Explained

Resource pooling is the mechanism by which a cloud provider serves many customers simultaneously from a shared infrastructure. The customer generally has no control or knowledge over the exact physical location of the provided resources but may be able to specify location at a higher level of abstraction (e.g., country, state, or datacenter).

- **Multi-tenancy:** A single physical server may run multiple virtual machines belonging to different customers simultaneously.
- **Shared infrastructure:** Resources (CPU, memory, storage, bandwidth) are drawn from a common pool rather than dedicated hardware.
- **Location independence:** Physical resources are abstracted using virtualization (hypervisors, SDN, SDS).
- **Dynamic assignment:** Resources are dynamically allocated and de-allocated based on current demand.
- **Security isolation:** Each tenant's data and workload is logically isolated from others despite sharing physical hardware.

Economies of Scale: By pooling resources, cloud providers achieve significant economies of scale, which translates to lower cost per unit of compute, storage, or bandwidth compared to individual enterprise deployments.

4.RAPID ELASTICITY:

Capabilities can be elastically provisioned and released, in some cases automatically, to scale rapidly outward and inward commensurate with demand. To the consumer, the capabilities available for provisioning often appear to be unlimited and can be appropriated in any quantity at any time.

2.4 Rapid Elasticity – Explained

Elasticity is the ability of the cloud to automatically scale resources up (scale-out) or down (scale-in) in response to workload fluctuations. This is fundamentally different from traditional capacity planning, where organisations had to purchase hardware based on estimated peak loads — leading to either over-provisioning (wasted cost) or under-provisioning (poor performance).

- **Auto-scaling:** Applications automatically add server instances when traffic spikes.
- **Scale-in:** Resources are released when demand drops, reducing unnecessary cost.
- **Perceived infinity:** From the user's perspective, the pool of available resources appears virtually infinite.
- **Policy-driven:** Elasticity can be triggered by metrics like CPU utilisation, request count, memory usage, or queue depth.

⚡ ELASTICITY vs SCALABILITY

Aspect	Elasticity	Scalability
Direction	Both scale-out and scale-in	Primarily scale-up or scale-out
Automation	Automatic, real-time	Often manual or planned
Duration	Short-term (minutes to hours)	Long-term capacity planning
Cost Impact	Pay only for what is used	May require pre-purchase
Example	AWS Auto Scaling Group	Upgrading DB to larger instance

5. MEASURED SERVICE:

Cloud systems automatically control and optimize resource use by leveraging a metering capability at some level of abstraction appropriate to the type of service (e.g., storage, processing, bandwidth, and active user accounts). Resource usage can be monitored, controlled,

and reported — providing transparency for both the provider and consumer of the utilized service.

2.5 Measured Service – Explained

Also referred to as the pay-per-use or metered model, this characteristic ensures that cloud resource consumption is measurable, transparent, and billed accordingly. It replaces the traditional model of fixed capital expenditure (CapEx) with a variable operational expenditure (OpEx) model.

- **Granular metering:** Every unit of resource (CPU hours, GB-months of storage, API calls, data transfer) is metered precisely.
- **Pay-as-you-go billing:** Users are charged only for what they actually consume — no more, no less.
- **Usage transparency:** Dashboards show real-time and historical usage across all resource types.
- **Cost control:** Budget alerts and spending caps can be configured to prevent unexpected charges.
- **Accountability:** Both provider and consumer gain visibility into resource utilisation.

Examples of Metered Resources in Major Clouds:

Resource Type	AWS Metric	Azure Metric	GCP Metric
Compute	EC2 instance-hours	VM hours (vCore)	Compute Engine vCPU-hours
Storage	S3 GB-month	Blob Storage GB-month	Cloud Storage GB-month
Network	Data transfer GB	Outbound bandwidth GB	Egress GB
Database	RDS instance-hours + I/O	Azure SQL DTU-hours	Cloud SQL instance-hours

CONSOLIDATED SUMMARY OF ALL FIVE CHARACTERISTICS :

Characteristic	Core Concept	Business Impact
On-Demand Self-Service	Automated, instant provisioning without human interaction	Faster time-to-market; reduced IT overhead

Broad Network Access	Access from any device over standard networks	Increased mobility and user flexibility
Resource Pooling	Shared multi-tenant infrastructure with dynamic allocation	Cost efficiency through economies of scale
Rapid Elasticity	Automatic scale-up and scale-down based on demand	Match capacity to load; avoid waste
Measured Service	Metered, transparent, pay-per-use billing	Shift CapEx to OpEx; financial predictability

INTERRELATION AMONG CHARACTERISTICS:

The five essential characteristics are not independent; they are deeply interconnected and collectively deliver the cloud computing value proposition:

- Resource Pooling creates the large shared infrastructure that makes elasticity and self-service feasible at scale.
- Rapid Elasticity relies on the pooled resources to provision and de-provision instantly in response to demand.
- On-Demand Self-Service gives the consumer control over elasticity without requiring provider intervention.
- Broad Network Access ensures that self-provisioned, elastic resources are reachable from anywhere.
- Measured Service provides the accountability layer — tracking exactly how pooled, elastic resources are consumed — and drives the pay-as-you-go billing model.

Without all five working together, a service cannot be classified as true cloud computing under the NIST framework.

TRADITIONAL COMPUTING VS. CLOUD COMPUTING:

Parameter	Traditional Computing	Cloud Computing
Provisioning	Days to weeks (manual process)	Minutes to seconds (automated)

Scalability	Limited by physical hardware; requires procurement	Virtually unlimited; on-the-fly scaling
Cost Model	CapEx (upfront purchase & depreciation)	OpEx (pay-as-you-go)
Access	LAN/VPN; location-dependent	Internet; location-independent
Maintenance	Managed by in-house IT team	Managed by cloud provider (shared model)
Disaster Recovery	Expensive; requires duplicate hardware	Built-in redundancy; geo-replication available
Resource Utilisation	Typically 10–20% of capacity used	High utilisation through multi-tenancy

REAL-WORLD APPLICATIONS OF EACH CHARACTERISTIC:

The following scenario illustrates all five characteristics in a single real-world case study:

 CASE STUDY: E-Commerce Website during a Flash Sale

Scenario: An e-commerce company hosts its website on AWS and runs a 24-hour flash sale that generates 10× normal traffic.

On-Demand Self-Service: The DevOps team pre-configures auto scaling policies via the AWS Console — no manual server requests.

Broad Network Access: Customers access the sale from mobile phones, browsers, and apps worldwide over HTTPS.

Resource Pooling: AWS allocates additional EC2 instances from its shared regional pool to meet demand.

Rapid Elasticity: During the sale, the auto scaling group grows from 4 to 40 instances within minutes; after the sale, it scales back to 4.

Measured Service: AWS bills only for the 36 additional instance-hours consumed during the sale; the company pays ~\$14.40 extra instead of buying 36 permanent servers.

KEY TERMS & DEFINITIONS:

Term	Definition
On-Demand Self-Service	Ability to provision resources automatically without human intervention from the provider.
Broad Network Access	Availability of cloud services over the network via standard protocols from heterogeneous devices.
Resource Pooling	Multi-tenant sharing of physical resources dynamically assigned to multiple consumers.
Multi-tenancy	Architecture where multiple customers share the same physical infrastructure with logical isolation.
Rapid Elasticity	The ability to automatically scale resources in and out in response to demand.
Measured Service	Automatic monitoring, control, and reporting of resource usage enabling pay-per-use billing.
CapEx	Capital Expenditure – upfront investment in hardware and infrastructure.
OpEx	Operational Expenditure – ongoing costs for services consumed, as in the cloud model.
Auto Scaling	Cloud feature that automatically adjusts the number of compute resources based on defined metrics.
Hypervisor	Software layer that enables multiple virtual machines to run on a single physical server.